

University: The University of Hong Kong (HKU)
Faculty: Engineering
Title of case study: UTEE: A Secure, Efficient, and Portable Distributed Bigdata Computing System on Heterogeneous Trusted Execution Devices
1. Summary of the impact (indicative maximum 100 words) We developed UTEE (Ubiquitous Trusted Execution Environments), the core software system of Huawei's secure cloud service (www.huaweicloud.com/product/tics.html), for public cloud users to conduct bigdata computing securely. Specifically, our novel techniques in UTEE can enforce data confidentiality and integrity strictly (e.g., the decrypted data being processed on the cloud cannot be peaked or tampered by even the malicious administrators of the cloud). Among relevant academic or commercial competitor systems in the world, UTEE has four unique features: (1) easy-to-program (Java-compatible), (2) bigdata-friendly (bigdata dataset compatible), (3) CPU-and-GPU-compatible, and (4) highest performance. Dr. Cui's UTEE research team at HKU has made diverse impacts beyond academia, including numerous media reports, four HK government grants (three GRF and one ITF), two industrial-oriented technology-transfer grants from Huawei, the submitted applications of three USA patents, and one HKU Faculty Knowledge Exchange Award. Huawei's official appreciation letter (for short, 'Huawei letter') attached at the end of this document reveals that, UTEE is already being used by Huawei Cloud's 3 million customers (enterprises and individuals) in 170 countries. 2. Underpinning research (indicative maximum 500 words) [Background of TEE] In this bigdata era, people usually conduct computations on sensitive user data (e.g., decrypted data) using bigdata computing software (e.g., Spark and Flink), and the computation tasks of each bigdata software are spitted and ran on a massive number of (heterogeneous) CPUs and GPUs in a public cloud (e.g., Amazon Cloud, Huawei Cloud, and Hospitals that conduct big-data computations). In recent years, a powerful and hardware-enforced security technique called TEE (Trusted Execution Environments), is introduced in high-end commercial CPU processors (e.g., Intel SGX and Arm TrustZone). TEE is a processor's special, tiny 'execution zone', which is physically isolated from the rest of the processor. When a bigdata task runs in a TEE, TEE's hardware protects the confidentiality and integrity of the task and data, strictly against the entire cloud software and people who have high software privileges, including malicious/flawed software (e.g., OS kernels with bugs) and malicious cloud administrators. Unfortunately, despite much effort from worldwide academia and industrial competitor systems (including Microsoft's VC3, European Union's SGX-Spark, UC Berkeley's Opaque, Google's Confidential Computing, and UNC's Civet), all existing TEE-relevant systems still faces three significant open research problems: (1) confirmed by all these competitor systems, TEE is still a low-level (assembly or C languages) programming object, so it is extremely hard to use by regular cloud software developers; (2) confirmed by our [Cronous ACM Micro 2022] paper in Section 3, TEE is only commercially available in some CPUs, but not in general GPUs; (3) revealed by our [Uranus ACM AsiaCCS 2020] paper in Section 3, existing competitors' TEE systems suffer from 20X~100X performance overhead compared to the native (insecure) bigdata computations, and cannot support typical big-data datasets (e.g., terabyte). To tackle these open problems, Dr' Cui's HKU team developed novel security techniques and integrate them as a new system called UTEE (Ubiquitous Trusted Execution Environments), including notable features below: [1. Easy-to-program] UTEE is one of the world's first industrial-grade systems that support invoking unmodified bigdata queries (written in Java) in a TEE's isolated execution zone, including two pieces of impact evidence. First, our [Uranus ACM AsiaCCS 2020] paper takes the

first step among all the published papers/reports about revealing that, all existing competitor systems before our paper's publication in Oct 2020, require intense and manual modifications on the bigdata queries (or even sometimes rewriting the entire bigdata queries from Java to C), in order to run these queries in any TEE-related processor. Second, the 1st paragraph of **Huawei letter** explicitly confirms that, 'UTEE (mainly, developed and patented from our Uranus paper) is a novel (i.e., first project in global industry) system combining the complementary advantages of the (TEE) hardware components and high-level programming languages (e.g., Java and Scala)'.

[2. Smallest trusted computing base (TCB)] In any security system, TCB means the set of all software and hardware components that must be assumed to be fully trusted and secure of the system; the small the TCB of a system, the more secure the system is. Our UTEE system has the smallest TCB among the competitor systems, with three pieces of impact evidence. First, claimed in our [Uranus AsiaCCS] paper, because UTEE is the world's first work to support 'running only the bigdata queries (including all the decrypted data and sub-functions) within a CPU processor secure execution zone', and the rest of the cloud software (including OS and the big-data computing software which invoke the bigdata queries), are all made by UTEE automatically to run outside the execution zone, UTEE's TCB is around millions of lines of code smaller than all competitor systems. Second, explicitly quoted in the 3rd paragraph of Huawei letter, Huawei appreciated several novel techniques in UTEE, especially its 'low TCB', and so it is 'suitable to deploy on clouds, but also on personal computers and mobile phones'. Third, claimed in our [Cronus ACM Micro 2022] paper, UTEE requires only the CPU processors to contain TEE hardware, and UTEE is the first work to enable software-style-TEE in general GPUs (i.e., these GPUs' hardware do not need to support TEE). These three reasons make UTEE a unique system to support almost all world's general GPU brands, and UTEE being usable by Huawei's 3 million customers in 170 countries (this significant user base is confirmed in the third paragraph of Huawei letter). Fourth, government examinations (XXX, Section 5).

[3. Big-data friendly and Highest performance] Two pieces of impact evidence exist. First, the concrete evaluation results of our [Uranus AsiaCCS] paper claimed two facts:: (a) UTEE is world's first system that enables the TEE-enabled secure executions on typical bigdata datasets (e.g., terabyte on some typical bigdata queries and algorithms); (b) this size of dataset being able to run in UTEE (Uranus) is two or three orders of magnitudes larger than those datasets being able to run by competitor systems'. Second, praised in the 3rd paragraph of Huawei letter, Huawei explicitly praised that, UTEE has the novel technique of 'big-data aware high-performance processing'.

3. References to the research (indicative maximum of six references)

Published international top tier papers (all papers are **listed as top venues by Google Scholar** in the Computer Security field

https://scholar.google.com.hk/citations?view_op=top_venues&hl=en&vq=eng_computersecuritycryptography or the Computing Systems field

https://scholar.google.com/citations?view_op=top_venues&hl=en&vq=eng_computingsystems).

All the first authors of these papers below are Dr. Heming Cui's PhD students. "*" indicates the corresponding author.

- (1) Ji Qi, Xusheng Chen, Yunpeng Jiang, Jianyu Jiang, Tianxiang Shen, Shixiong Zhao, Sen Wang, Gong Zhang, Li Chen, Man Ho Au, and **Heming Cui***, BIDL: A High-throughput, Low-latency Permissioned Blockchain Framework for Datacenter Networks, Proceedings of the 28th ACM Symposium on Operating Systems Principles (SOSP '21). **ACM open source and results reproduced badge.**

- (2) Jianyu Jiang, Qi Ji, Tianxiang Shen, Xusheng Chen, Shixiong Zhao, Sen Wang, Li Chen, Gong Zhang, Xiapu Luo, and **Heming Cui***, CRONUS: Fault-isolated, Secure and High-performance Heterogeneous Computing for Trusted Execution Environments, Proceedings of the 55th ACM/IEEE International Symposium on Microarchitecture (**MICRO '22**). **ACM open source and results reproduced badge**.
- (3) Shixiong Zhao, Fanxin Li, Xusheng Chen, Tianxiang Shen, Li Chen, Sen Wang, Gong Zhang, Cheng Li, and **Heming Cui***, NASPipe: High Performance and Reproducible Pipeline Parallel Supernet Training via Causal Synchronous Parallel, the 2022 Conference on Architectural Support for Programming Languages and Operating Systems (**ASPLOS '22**). **ACM open source and results reproduced badge**.
- (4) Tianxiang Shen, Jianyu Jiang, Yunpeng Jiang, Xusheng Chen, Ji Qi, Shixiong Zhao, Fengwei Zhang, X. Luo, and **Heming Cui***, Making Strong Anonymity Scale in a Fully Decentralized Network, IEEE Transactions on Dependable and Secure Computing 2021 (**TDSC '21**).
- (5) Shixiong Zhao, Fanxin Li, Xusheng Chen, Xiuxian Guan, Jianyu Jiang, Dong Huang, Yuhao Qing, Sen Wang, Peng Wang, Gong Zhang, Cheng Li, Ping Luo, and **Heming Cui***, vPipe: A Virtualized Acceleration System for Achieving Efficient and Scalable Pipeline Parallel DNN Training, IEEE Transactions on Parallel and Distributed Systems 2021 (**TPDS '21**).
- (6) Jianyu Jiang, Xusheng Chen, Tzs On Li, Cheng Wang, Tianxiang Shen, Shixiong Zhao, **Heming Cui***, Co-Li Wang, and Fengwei Zhang, Uranus: Simple, Efficient SGX Programming and Its Applications, Proceedings of the 15th ACM ASIA Conference on Computer and Communications Security (**ASIACCS '20**).

4. Details of the impact (indicative maximum 750 words)

The achieved real-world impact of this UTEE project **after July 2022 (the Faculty KE Award)**:

- (1) Dr. Cui's RGC RIF proposal (Ref. R7030-22F, with Dr. Cui the PC and a budget of HK \$4.3 million), titled "MindPipe: High-performance and Carbon-efficient Four-dimensional Parallel Training System for Large AI Models", is **shortlisted for the final-round interview with the RIF panel** (Dr. Cui, the PC of the RIF proposal, will attend the interview in Dec 7, 2022).
- (2) **Both Huawei** (<https://hemingcui.github.io/doc/huawei-stamped-letter.pdf>) **and SenseTime** (<https://hemingcui.github.io/doc/sensetime-letter.pdf>) provided explicit support letters for Dr. Cui's MindPipe proposal, with explicit interests stated in the letters about deploying MindPipe.
- (3) The MindPipe training system (or also called the "Fold4D" system, interchangeably) has been deployed in Huawei's supercomputing datacenter and achieved even faster training performance than USA's fastest relevant training system (the Megatron system, <https://www.marktechpost.com/2021/04/19/researchers-from-nvidia-stanford-university-and-microsoft-research-propose-efficient-trillion-parameter-language-model-training-on-gpu-clusters/>, built by Stanford, Microsoft and Nvidia). So, Dr. Cui's MindPipe (or "Fold4D") system **received a Best Collaborating Deliverables Award from Huawei in Oct 2022** (<https://hemingcui.github.io/doc/fold4d-award-2022.jpg>). Dr. Cui and Huawei **plan to announce the commercialization of the MindPipe system as Huawei's keystone training system for big AI models in a few months**.

The achieved real-world impact of this UTEE project **before July 2022 (the Faculty KE Award)**:

- (4) A preliminary prototype of UTEE (Ubiquitous Trusted Execution Environments), the proposed secure big-data system project, has commercially become the core system of Huawei Cloud's TICS (<https://www.huaweicloud.com/product/tics.html>).
- (5) Huawei's formal acknowledgement letter about the commercialization of UTEE (<https://hemingcui.github.io/doc/utee-ack.pdf>). This pdf letter comes with Huawei's corporate chop and its content is legally effective. This letter confirmed that, UTEE is already usable by Huawei Cloud's 3 million customers (including enterprises and individuals) in 170 countries.
- (6) Dr. Cui won the "**Best Collaborating Scientist Award Medal 2021**" of the Huawei Theory Research Department, who selects this award for the most-productive professor among their university collaborators around the world, including Europe/Canada/Asia (probably excluding

USA/UK since 2019 due to diplomatic reasons). See the award notification email (<https://hemingcui.github.io/doc/2021-award.pdf>) on why the award goes to Dr. Cui; and see the picture of the medal in the ppt slides.

- (7) **ITF ITSP Platform grant award** (an award letter is sent to HKU RSS and CS in May 2022, ref: GHP/169/20SZ). PI is Dr. Heming Cui. Grant title is “UTEE: A Secure, Efficient, and Portable Distributed Bigdata Computing System on Heterogeneous Trusted Execution Devices”, the same as the title of this KE Award. Grant amount is HK \$3.3 million, including 10% of the amount donated from Huawei. Grant period is June 2022 to May 2024.
- (8) Dr. Cui’s PhD students filed 13 patents and submitted to CINPA or WIPO (PCT).
- (9) **HKU-Med** (Dr. Teng Zhang) is collaborating with Dr. Cui to run UTEE for secure processing on patient data across the computers of the Queens Mary Hospital and HKU Medical School.

5. Sources to corroborate the impact (indicative maximum of 10 references)

In addition to the above external links/sources (including software releases, dissemination of usage across 3 million customers in 170 countries, Huawei’s best collaborating scientist award, and new ITF grant) to corroborate the impact in this word doc, there are also several external media reports to corroborate the impact of UTEE (TICS):

- (1) HDC.Cloud 2021: “Huawei Releases Six Groundbreaking Products to Supercharge the Cloud and Intelligent Transformation of Business” (<https://www.huawei.com/en/news/2021/4/hdc-cloud-2021-six-products-cloud-intelligent-transformation>).
- (2) Global Times: “Huawei launches six new products to bolster growth in cloud sector” (<https://www.globaltimes.cn/page/202104/1222038.shtml>).
- (3) China Daily (中国日报): “华为云可信智能计算服务 TICS，安全释放数据价值” (<https://caijing.chinadaily.com.cn/a/202104/29/WS608a7f1ea3101e7ce974cf35.html>).
- (4) 中国证券报官方网站: “第三届上海金融科技国际论坛：强化数据安全 释放融合价值” (https://www.cs.com.cn/xwzx/hg/202112/t20211205_6225416.html).
- (5) China government testing and review report: “12 月 18 日，2020 数据资产大会在北京召开。会上，中国信息通信研究院为通过“基于可信执行环境的数据计算平台”的产品颁发证书。华为云可信智能计算服务 TICS 基于鲲鹏 TrustZone 机密计算，结合硬件 TEE 和软件 SMPC 算法，实现了软硬结合的计算加速，同时支持跨信任域的联邦 SQL 分析和联邦学习能力，以九大测试项全部通过，树立隐私计算产品的新标杆。” (<https://icode.best/i/87602637140406>).
- (6) The TelecomReview.com: “Huawei launches 6 new products to power up Cloud business. Trusted Intelligent Computing Service (TICS): HUAWEI CLOUD TICS provides security for innovative and converged applications of data elements, enabling secure data flow and maximizing data value.” (<https://www.telecomreview.com/articles/telecom-vendors/4880-huawei-launches-6-new-products-to-power-up-cloud-business>).
- (7) The Min.news (“頭條匯- 匯聚天下頭條”): “HUAWEI CLOUD Trusted Intelligent Computing Service TICS, Safely Unleash the Value of Data” (<https://min.news/en/tech/c1d269423d6459b3c61e4391179224e3.html>).

Huawei’s formal acknowledgement letter about the commercialization of UTEE (<https://hemingcui.github.io/doc/utee-ack.pdf>) is also attached in the next page of this form.

Letter of Appreciation for Contribution to Technical Cooperation Project

To whom it may concern,

I, on behalf of the Theory Lab, Huawei Technologies Co., Ltd, am writing to show our commercialization acknowledgement of UTEE (Ubiquitous Trusted Execution Environments): a secure, efficient, and portable distributed big-data computing system based on advanced CPU secure hardware components (e.g., ARM TrustZone and Intel SGX). UTEE is a novel eco-system combining the complementary security advantages of these hardware components and high-level programming languages (e.g., Java and Scala). UTEE can effectively prevent the computations and queries on plaintext data being peeked by diverse malicious parties on public clouds, including operating systems, hypervisors, and malicious/careless cloud administrators.

My research team in the Huawei Theory Lab has been looking for solutions of secure big-data computing systems built by world-wide leading universities and industries. In 2018, we offered a two-year flagship research fund of HK \$2,189,450 to Dr. Heming Cui on jointly developing such systems. From this research fund, we have conducted substantial collaborations with Dr. Cui and produced diverse research achievements, including jointly publishing a series of research papers in international top journals/conferences (e.g., TDSC, TPDS, EuroSys, DSN, SRDS, and Asia CCS) and applying for seven patents. Below are three selected papers and two selected patents relevant to secure big-data computing systems. UTEE is most relevant to the last patent.

- Uranus: Simple, Efficient SGX Programming and Its Applications, 15th ACM ASIA Conference on Computer and Communications Security (ASIACCS 2020).
- DAENet: Making Strong Anonymity Scale in a Fully Decentralized Network, IEEE Transactions on Dependable and Secure Computing (TDSC 2021).
- UPA: An Automated, Accurate and Efficient Differentially Private Big-data Mining System, 50th IEEE/IFIP International Conference on Dependable Systems and Networks (DSN 2020).
- An Automated, Accurate and Efficient Differentially Private Big-data Mining System, submitted to CNIPA, Ref. CN 202010506698.X.
- An Efficient, Secure Big-data Processing and Programming System based on Trusted Execution Environment, submitted to CNIPA, Ref. CN 202010366539.4.

UTEE is one major resultant system developed from our collaborations. Compared to UTEE's worldwide competitors (e.g., Alibaba, Baidu and Microsoft), we appreciate several unique novel techniques in UTEE, including its low trusted computing base, big-data aware high-performance processing, and portability across heterogenous CPU vendors. These unique techniques of UTEE make it not only be suitable to deploy on clouds, but also on personal computers and mobile phones. In April 2021, our Huawei Clouds CEO announced the release of Trusted and Intelligent Cloud Services (TICS, see <https://www.huaweicloud.com/product/tics.html>), and UTEE has been successfully deployed in TICS as a core system component. This means that UTEE is usable by the 3 million Huawei Clouds customers (enterprises and individuals) located in 170 countries.

Overall, we believe the recognition of Dr. Cui's achievements on your side will be a major encouragement of his future research on developing secure systems that are both academically innovative and commercially practical. These developments can contribute to the effective protection of data for many entities, including individuals, enterprises, organizations, and governments.

Bo Bai
Sincerely,

Position / title: Director of the Theory Lab, Huawei Technologies Co., Ltd

Address: the Theory Lab, Administration Building Huawei Technologies Co., Ltd. Bantian, Longgang District, Shenzhen, P. R. China

Phone: 852-35472722

